Design and Simulation of a Digital Data Transmission Error Detection System Using XOR

Detecting errors | Ensuring reliable communication | Using the power of XOR



1. Introduction

In digital communication, data can get corrupted during transmission due to noise, interference, or other channel errors. Error detection techniques help identify these errors and ensure reliable data transfer. This project implements a simple and efficient error detection system using the **XOR (Exclusive OR)** operation.

2. Key Concepts

XOR (Exclusive OR)

- Outputs 1 only when the inputs are different.
- Used here to generate a checksum (parity).

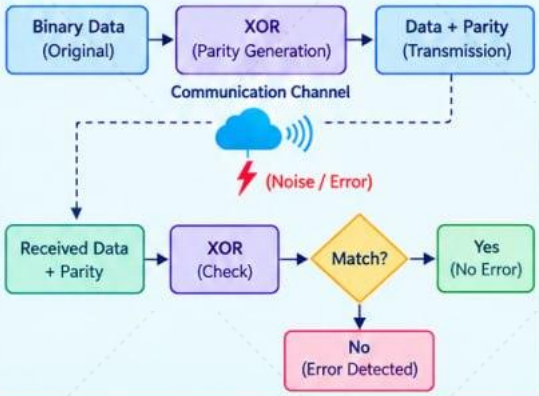


XOR Truth Table		
A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

Error Detection Principle

- The sender calculates a parity (XOR of all bits).
- The receiver calculates the parity again.
- If both match → No error.
- If they differ → Error detected.

3. System Design



4. Example

Let the data to be sent be: 1011001

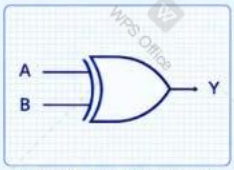
1. Calculate parity (XOR of all bits):
   1 ⊕ 0 ⊕ 1 ⊕ 1 ⊕ 0 ⊕ 0 ⊕ 1 = **0**
   (Even number of 1s → parity = 0)
2. **Transmit: 1011001 0**
3. If a bit flips during transmission (e.g., **1011101** 0), the receiver's XOR result will not be 0 → **Error detected.**

5. Simulation

The system was designed and simulated using Digital Logic Design tools (e.g., Logisim / Multisim).

Simulation Steps:

1. Enter the data bits.
2. Generate parity using XOR gate(s).
3. Simulate transmission (introduce error in one bit).
4. At the receiver, perform XOR check.
5. Display result: "No Error" or "Error Detected".



Simulation Circuit (Example)



Waveform Output

6. Results

Condition	Transmitted Data	Receiver Check	Result
No Error	10110010	0	No Error
With Error	10111010	1	Error Detected

7. Advantages

- Simple and easy to implement.
- **Low hardware** and computational cost.
- Effective for single-bit error detection.
- Suitable for real-time and low-complexity systems.

8. Conclusion

The XOR-based error detection system provides a simple, reliable and efficient method to detect errors in digital data transmission. The simulation confirms that the system successfully identifies corrupted data, making it suitable for real-world communication systems.

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